Assignment 6 -solns

1. A particle of charge e is constrained to move in a circle of radius R. The wavefunction, single-valued, depends only on the angle θ. The particle is acted on by a uniform electric field, $\vec{\mathcal{E}}$. Use effect of the field as a perturbation.

(i) find the unperturbed energy levels. What is the degeneracy of each level?

(ii) compute the level shifts through second order in H'.

Ans. $H_0 = -\frac{\hbar^2}{2I}\frac{\partial^2}{\partial\theta^2}, \quad H' = -e\mathcal{E}R\cos\theta$

H is even under $\theta \leftrightarrow -\theta$, hence we can take eigenstates to have definite parity, even (e) or odd (o) under $\theta \leftrightarrow -\theta$,

(i) For $H_0 u = E^{(0)} u, \quad E_n^{(0)} = \frac{n^2\hbar^2}{2I};\ n = 0,\ 1,\ 2, \cdots$

For $n = 0$, no degeneracy: $u_0(\theta) = \frac{1}{\sqrt{2\pi}}$

For $n > 0$, two fold degeneracy: $u_n^{(\mathrm{e})}(\theta) = \frac{1}{\sqrt{\pi}}\cos n\theta; \quad u_n^{(\mathrm{o})}(\theta) = \frac{1}{\sqrt{\pi}}\sin n\theta$;However, we do not really need to use methods of degenerate perturbation theory, because H' cannot mix the two degenerate states (see observation (1) below.

Observe: (a) $H'^{(\mathrm{e,\,o})}_{mn} = \langle u_m^{(\mathrm{e})}|H'|u_n^{(o)}\rangle = -\frac{e\mathcal{E}R}{\sqrt{\pi}}\cdot\frac{1}{\sqrt{\pi}}\int_0^{2\pi}\cos n\theta\cos\theta\sin n\theta d\theta = 0 = H'^{(\mathrm{o,\,e})}_{mn}$;

(b) $H'^{(\mathrm{e,\,e})}_{0,\,n} = \langle u_0|H'|u_n^{(\mathrm{e})}\rangle = H'^{(\mathrm{e,\,e})}_{n,\,0} = -\frac{e\mathcal{E}R}{\sqrt{2\pi}}\cdot\frac{1}{\sqrt{\pi}}\int_0^{2\pi}\cos n\theta\cos\theta d\theta = -\frac{1}{\sqrt{2}}eR\mathcal{E}\delta_{n,\,1}$

(c) $H'^{(\mathrm{o,\,o})}_{0,\,n} = \langle u_0|H'|u_n^{(\mathrm{o})}\rangle = H'^{(\mathrm{o,\,o})}_{n,\,0} = -\frac{e\mathcal{E}R}{\sqrt{2\pi}}\cdot\frac{1}{\sqrt{\pi}}\int_0^{2\pi}\sin n\theta\cos\theta d\theta = 0$

(d) For $m,\ n \geq 1$,

$$H'^{(\mathrm{e,\,e})}_{m,\,n} = H'^{(\mathrm{o,\,o})}_{m,\,n} = -\frac{e\mathcal{E}R}{\sqrt{\pi}}\cdot\frac{1}{\sqrt{\pi}}\int_0^{2\pi}\cos n\theta\cos\theta\cos m\theta d\theta = -\frac{e\mathcal{E}R}{\sqrt{\pi}}\cdot\frac{1}{\sqrt{\pi}}\int_0^{2\pi}\sin n\theta\cos\theta\sin m\theta d\theta = \begin{cases} \frac{-eR\mathcal{E}}{2}; & \forall\, m = n\pm 1 \\ 0 \ \text{otherwise} \end{cases}$$

Perturbation:

$\because H'^{(\mathrm{e,\,e})}_{n,\,n} = H'^{(\mathrm{o,\,o})}_{n,\,n} = -\frac{e\mathcal{E}R}{\sqrt{\pi}}\cdot\frac{1}{\sqrt{\pi}}\int_0^{2\pi}\cos^2 n\theta\cos\theta d\theta = -\frac{e\mathcal{E}R}{\sqrt{\pi}}\cdot\frac{1}{\sqrt{\pi}}\int_0^{2\pi}\sin^2 n\theta\cos\theta d\theta = 0\ \forall\, n$, $\therefore$there are no first order shifts.

To second order, $\Delta E_n^{(e)} = \sum_{m\neq n}\frac{\left|H'^{\mathrm{e\,e}}_{m\,n}\right|^2}{E_n^{(0)}-E_m^{(0)}} = \frac{\left|H'^{\mathrm{e\,e}}_{n+1\,n}\right|^2}{E_n^{(0)}-E_{n+1}^{(0)}} + \frac{\left|H'^{\mathrm{e\,e}}_{n-1\,n}\right|^2}{E_n^{(0)}-E_{n-1}^{(0)}}$

and similar formula for odd one.

(a) $n > 1, \quad \Delta E_n^{(\mathrm{e})} = \Delta E_n^{(o)} = \frac{2e^2\mathcal{E}^2R^2I}{4\hbar^2}\left\{\frac{1}{n^2-(n+1)^2} + \frac{1}{n^2-(n-1)^2}\right\} = \frac{e^2\mathcal{E}^2R^2I}{\hbar^2}\frac{1}{4n^2-1}$

(b) $n = 1, \quad \Delta E_1^{(\mathrm{e})} = \frac{\left|H'^{e\,e}_{0\,1}\right|^2}{E_1^{(0)}-E_0^{(0)}} + \frac{\left|H'^{e\,e}_{2\,1}\right|^2}{E_1^{(0)}-E_2^{(0)}} = \frac{5e^2\mathcal{E}^2R^2I}{6\hbar^2}$

$\Delta E_1^{(\mathrm{o})} = \frac{\left|H'^{\mathrm{o\,o}}_{2\,1}\right|^2}{E_1^{(0)}-E_2^{(0)}} = -\frac{e^2\mathcal{E}^2R^2I}{6\hbar^2}$

(c) $n = 0, \quad \Delta E_0^{(\mathrm{e})} = \frac{\left|H'^{\mathrm{e\,e}}_{1\,0}\right|^2}{E_0^{(0)}-E_1^{(0)}} = -\frac{e^2\mathcal{E}^2R^2I}{\hbar^2}$

2. Show that the minimum uncertainty wave packet is a gaussian.

Ans. see attached Griffiths p.108-109

3. Use the generalised uncertainty principle, $\langle\hat{A}^2\rangle\langle\hat{B}^2\rangle \geq \frac{1}{4}\langle i[\hat{A},\hat{B}]\rangle^2$, with $A \equiv H$. Define $\Delta E \equiv \sigma_H$ and $\Delta t \equiv \frac{\sigma_B}{|\frac{d\langle B\rangle}{dt}|}$. Show that $\Delta E\Delta t \geq \frac{\hbar}{2}$

Ans. see attached Griffiths p.109-111

4. A particle moves in a 1-D space, in a potential $V(x)$ which vanishes as $x \to \pm\infty$. In at least some range of x, $V(x) < 0$, and nowhere is it positive. Show that there is at least one bound state. Hint : Find a trial function (e.g., a gaussian), for which $\langle E \rangle < 0$.

Ans. Locate the origin so that V(x) is negative definite in some interval $-a < x <$a :

$-\int_{-a}^{a} V(x)dx \equiv -A(a)$ where A(a) > 0 .

For normalized trial function, take $\psi = \left(\frac{\lambda}{\pi}\right)^{\frac{1}{4}} e^{-\frac{\lambda x^2}{2}}$

$\langle \Psi|H|\Psi \rangle = \langle E \rangle = \frac{\hbar^2}{2m}\lambda + \sqrt{\frac{\lambda}{\pi}} \int_{-a}^{a} V(x) e^{-\lambda x^2} dx$

but $\int_{-a}^{a} dx\, V(x) e^{-\lambda x^2} \leq e^{-\lambda a^2} \int_{-a}^{a} dx\, V(x) = -e^{-\lambda a^2} A(a)$

$\langle E \rangle \leq \frac{\hbar^2}{2m}\lambda - \sqrt{\frac{\lambda}{\pi}} A(a) e^{-\lambda a^2}$

For small λ , the second term dominates, since $\sqrt{\lambda} e^{-\lambda a^2} > \lambda$.

Choosing λ small enough, we obtain a trial function such that $\langle E \rangle < 0$.

5. A particle moves in a 1-D space, in a potential, $V(x) = -\frac{\hbar^2}{2m}.\lambda\{\delta(x-b) + \delta(x+b)\}$. Discuss the bound state energy eigenvalue problem. Consider the even and odd parity eigenfunctions separately.

(i) Show that there is always one and only one even parity bound state. Find the eigenvalue for this state in the limits $b \to 0$ and $b \to \infty$.

(ii) For the odd parity case, show that there is only one bound state if λb is larger than a critical value and no bound state otherwise. What is the critical value of λb? Find the eigenvalues for $\lambda b \to \infty$.

Ans. Notation : $u(x)$ is the wavefunction. The hyperbolic functions $\cosh(x)$ and $\sinh(x)$ are sum and difference of $e^{\pm x}$ with appropriate coefficients.

$\frac{d^2u}{dx^2} - k^2 u = -\lambda\delta(x-b) + \delta(x+b)$;

$k = \sqrt{\frac{-2mE}{\hbar^2}}, \quad E = -\epsilon$

(a) Even parity case, $|x| < b, \quad u_{\rm I} = A\cosh kx$

$x > b, \quad u_{\rm II} = Be^{-kx}$

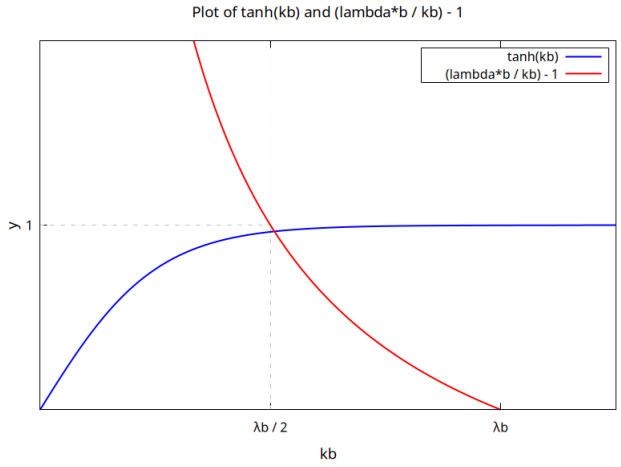
B.C. : at $x = b$, $u_{\rm II}(b) = u_{\rm I}(b) \equiv u(b)$

and $u'_{\rm II}(b) - u'_{\rm I}(b) = -\lambda u(b)$

From these, we get $A\cosh kb = Be^{-kb}$

and $\quad -kBe^{-kb} - kA\sinh kb = -\lambda Be^{-kb}$

$\Rightarrow \tanh kb = \frac{\lambda}{k} - 1 = \frac{\lambda b}{kb} - 1$



Clearly, there is one and only one solution for k, and $\frac{\lambda}{2} \leq k \leq \lambda$

when $b \to 0$, $k \to \lambda$

with $b \to 0$, we have only one well at $x = 0$ of strength 2λ.

and $k = \frac{2\lambda}{2} = \lambda$ from earlier results derived in class for a single well.

when $b \to \infty$, $\quad k \to \frac{\lambda}{2}$

(b) Odd parity case: $|x| < b$: $\quad u_{\mathrm{I}} = A \sinh kx$

$x > b$: $\quad u_{\mathrm{II}} = Be^{-kx}$

at $x = b$ using B.C. as before, $\tanh kb = \frac{1}{\frac{\lambda}{k} - 1}$

Trivial solution : k=0, but that gives $u_I = 0,\ u_{II} = 0$

Non-trivial solution: only if $\lambda b \geq 1$ as $b \to \infty,\ k \to \frac{\lambda}{2}$; $\quad$ as $\lambda b \to 1,\ k \to 0$